

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Analytical Problem Solving

Course Code:	ITA0517	Course Title:	Computer Vision
CO Assessed:	CO2 – Imitation (BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)		
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Section / Batch:	BE-CSE-AI/2024-2028	Date:	05-08-26

Code of Conduct

I Malli Chandana(192472250)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Malli Chandana

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Criteria	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	Problem Understanding	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	Method Selection	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	Solution Process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	Accuracy of Calculation	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	Interpretation of Results	

1. High-Pass Filtering An image requires sharpening.

(a) Interpret need for high-pass filtering.

(b) Justify filter selection.

(c) Apply mask:]

(d) Compute center value.

(e) Interpret edge enhancement.

(a) Interpret the need for high-pass filtering

High-pass filtering is used to **sharpen an image** by emphasizing high-frequency components such as edges, corners, and fine textures while reducing the influence of low-frequency components like smooth backgrounds and gradual intensity changes. It improves the visual quality of images by making object boundaries clearer and enhancing important details. This technique is widely used in medical imaging, satellite image processing, fingerprint recognition, and document enhancement.

(b) Justify filter selection

A commonly used high-pass filter (Laplacian mask) is:

Justification:

- Detects intensity changes in all directions.
- Enhances edges without depending on edge orientation.
- Suppresses uniform image regions.
- Produces a sharper and more detailed image.
- Frequently used for image enhancement before feature extraction.

(c) Apply the mask

Assume the 3×3 image window is:

Applying the filter:

(d) Compute the center value

Since an 8-bit image allows pixel values only between **0 and 255**, the computed value is clipped to **255**.

(e) Interpret edge enhancement

The high-pass filter enhances regions with sudden intensity changes while suppressing slowly varying regions. As a result:

- Object boundaries become sharper.
- Fine image details become more visible.
- Blurred edges are improved.
- Smooth areas remain dark or close to zero.
- Overall image sharpness and clarity increase.

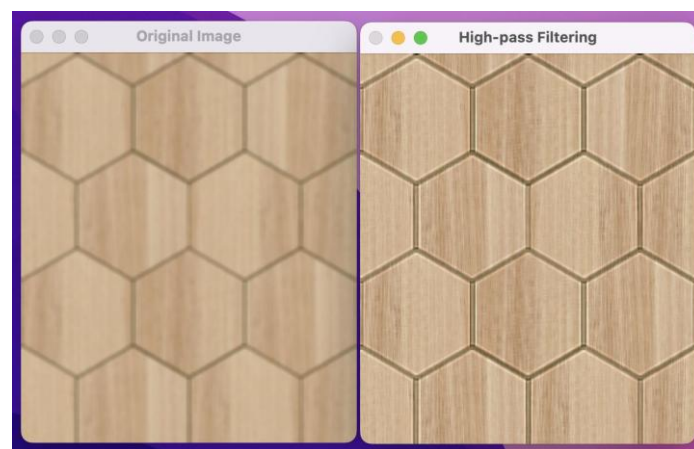


Fig:"Before vs After"

Observation:

- Before filtering: Image appears blurred with smooth edges.
- After filtering: Edges become sharper and fine details are more visible.

Applications:

- Medical image enhancement
- Satellite image sharpening
- Fingerprint recognition
- Face recognition
- Industrial quality inspection

17. Frequency Domain High-Pass

Given frequency components:

- (a) Interpret high-frequency importance.**
- (b) Justify high-pass filtering.**
- (c) Apply filtering (remove low frequencies).**
- (d) Compute output.**
- (e) Interpret result.**

Given frequency components:

where **100 and 80** represent low-frequency components and **40, 20, 10** represent high-frequency components.

(a) Interpret the importance of high-frequency components

High-frequency components represent rapid changes in image intensity. They contain important visual information such as:

- Edges
- Fine textures
- Small objects
- Surface details
- Object boundaries

Without these components, an image appears blurred and loses sharpness.

(b) Justify high-pass filtering

High-pass filtering removes low-frequency components that correspond to background illumination and smooth intensity variations. By retaining only the high-frequency components, the filter highlights edges and fine details, making the image more suitable for feature extraction, object recognition, and image analysis.

(c) Apply filtering (remove low frequencies)

Remove:

Remaining frequency components:

(d) Compute the output

Filtered output:

(e) Interpret the result

After filtering:

- Low-frequency background information is eliminated.
- High-frequency details are preserved.
- Edges become more prominent.
- Image sharpness increases.
- The processed image is better suited for edge detection and feature analysis.
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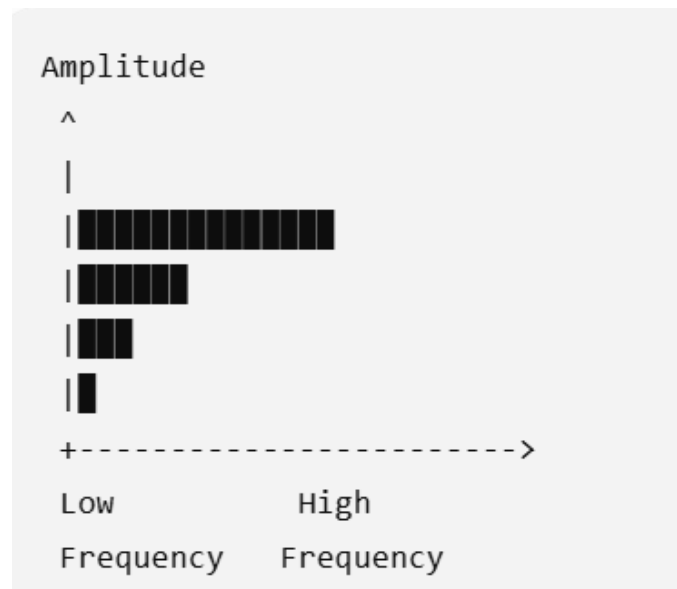


Fig:Frequency Spectrum Diagram

Before:				
100	80	40	20	10
After HPF:				
0	0	40	20	10

Applications

- Edge enhancement
- Texture analysis
- Image restoration
- Medical imaging
- Pattern recognition

18. Adaptive Thresholding

Pixel values: [60, 80, 120, 140]

- (a) Interpret segmentation challenge.**
- (b) Justify adaptive thresholding.**
- (c) Apply local threshold (mean = 100).**
- (d) Compute binary output.**
- (e) Interpret result.**

Given pixel values:

Local threshold:

Decision rule:

- $\text{Pixel} \geq 100 \rightarrow 1$ (Foreground/White)
- $\text{Pixel} < 100 \rightarrow 0$ (Background/Black)

(a) Interpret the segmentation challenge

Image segmentation becomes difficult when different parts of an image have varying illumination or shadows. A single global threshold cannot accurately separate foreground and background in such situations. Adaptive thresholding solves this problem by calculating a threshold based on the local neighborhood of each pixel, allowing accurate segmentation even under uneven lighting conditions.

(b) Justify adaptive thresholding

Adaptive thresholding is preferred because:

- It handles non-uniform illumination effectively.
- Produces more accurate segmentation than global thresholding.
- Preserves object boundaries.
- Reduces segmentation errors caused by shadows and brightness variations.
- Commonly used in document image processing, medical imaging, and license plate recognition.

(c) Apply the local threshold

Pixel Value	Threshold (100)	Binary Output
60	$60 < 100$	0
80	$80 < 100$	0
120	$120 \geq 100$	1
140	$140 \geq 100$	1

(d) Compute the binary output

[0,0,1,1]

(e) Interpret the result

The adaptive threshold successfully separates the foreground from the background based on local intensity values. Pixels with values **120 and 140** are identified as foreground (white), while **60 and 80** are classified as background (black). This method provides accurate segmentation even when image brightness changes across different regions, making it more reliable than a fixed global threshold.

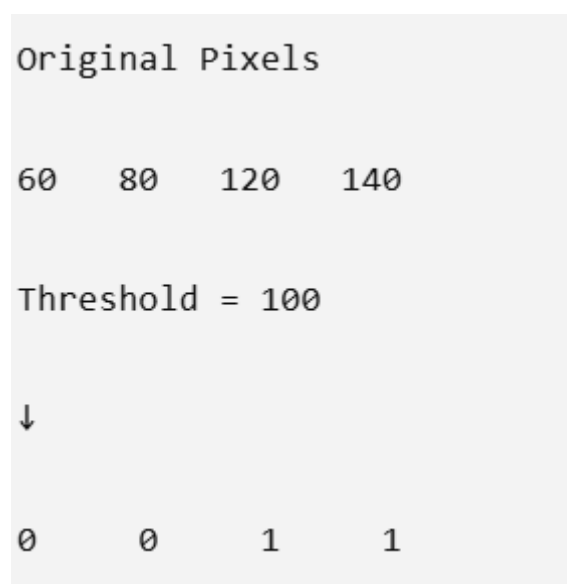


Fig:Simple Binary Conversion Diagram

Applications

- Document scanning
- OCR (Optical Character Recognition)
- Medical image segmentation
- Number plate detection
- Face detection